

Algebraicity & finite extensions

Proposition (1)

$$L/k, \alpha \in L$$

TFAE:

(1) α is algebraic over k

(2) $\ker \text{ev}_\alpha \neq (0)$

(3) $k[\alpha] = k(\alpha)$

(4) $k(\alpha)$ is a field.

(5) $k(\alpha)$ is a finite extension of k .

(6) There exists a sub-extension
 $k \subseteq M \subseteq L$ s.t. (i) M/k is finite
 $\downarrow$
 α

Remark: (1) M/k is not
a quotient

(ii) $\alpha \in M$

(2) Sub-extension $(\Leftrightarrow)$ M is a sub-field of L containing k .

Pf: Last time, we showed

$$(1) \Leftrightarrow (2) \Leftrightarrow (3) \Leftrightarrow (4).$$

$$(5) \Rightarrow (6) \quad \checkmark \quad \text{Take } M = k(\alpha).$$

$$(6) \Rightarrow (5) \quad k \subseteq k(\alpha) \subseteq M$$

M is a f.dim k -v. space
and $k(\alpha) \subseteq M$ is a subspace.

$\Rightarrow k(\alpha)$ is also f.dim'l as
a k -vector space

$\Rightarrow k(\alpha)/k$ is a finite extension

$$(4) \Rightarrow (6) : \text{When } k[\alpha] \text{ is a field}$$

$$\underbrace{k[x]/(m_\alpha(x))}_{\text{finite extension of } k} \xrightarrow[\text{ev}_\alpha]{\cong} k[\alpha]$$

$\Rightarrow k[\alpha]$ is a finite extⁿ of k
containing α .

$\Rightarrow M = k[\alpha]$ works.

(6) $\Rightarrow$ (1) :

Given: M/k finite & $\alpha \in M$

$$\dim_k M = d$$

$1, \alpha, \alpha^2, \dots, \alpha^d$ are linearly dependent
 $d+1$

$\Rightarrow \exists a_0, \dots, a_d \in k$ not all zero

s.t. $a_0 + a_1 \alpha + \dots + a_d \alpha^d = 0$

Then $f(x) = a_0 + a_1x + \dots + a_dx^d \in k[x] \setminus \{0\}$

Satisfies $f(\alpha) = \text{ev}_\alpha(f(x)) = 0$,

$\Rightarrow \alpha$ is algebraic over k

Example:

$$\begin{array}{c} \mathbb{F}_{125} = \mathbb{F}_{5^3} \\ \swarrow \quad \searrow \\ \mathbb{F}_5(\alpha) \quad | \quad \leftarrow \text{no fields in-between} \\ \swarrow \quad \searrow \\ \mathbb{F}_5 \end{array}$$

$$\mathbb{F}_5[\alpha] = \mathbb{F}_5[x] / (m_\alpha(x))$$

$$\alpha \in \mathbb{F}_{125} \setminus \mathbb{F}_5 \Rightarrow \mathbb{F}_5(\alpha) = \mathbb{F}_{125}$$

$$\Rightarrow \deg m_\alpha(x) = 3$$

There are 120 such α .

All such α generate $\mathbb{F}_{125}$ as an extension of $\mathbb{F}_5$.

We can also look at generators of $\mathbb{F}_{125}^\times$

as a cyclic group : There are 60 of these.

Observation ①

$L / M / k$: sequence of extensions



L

$|$

M

$|$

k

$(\Rightarrow) \quad k \leq M \leq L$

Suppose that L/M is finite

& M/k is finite

then L/k is finite

and $[L:k] = [L:M][M:k]$

Proposition (2)

L/k : extension
 α, β : algebraic / k

Then the following elements are all algebraic over k :

(i) $c \cdot \alpha \quad \forall c \in k$

(ii) $\alpha + \beta$

(iii) $\alpha \beta$

(iv) $(\alpha \neq 0) \alpha^{-1}$

Corollary: $M = \{ \alpha \in L : \alpha \text{ is algebraic over } k \}$

Then M is a sub-extension of L over k and is

the largest subextension consisting entirely of algebraic elements.

Defn An extension M/k is
algebraic if every $\alpha \in M$
is algebraic over k .

Observation (2) (6) $\Rightarrow$ (1) of

Proposition (1) tells us that
any finite extension is algebraic.